

Female Reproductive System

Laboratory Experience

The purpose of this laboratory exercise is to identify the different tissues and cells found in the female reproductive system. Identify the cell types present and the tissue structure in each of the different organs. Use the slides listed below along with your manual, your text and your atlas to identify the cells, tissues and organs described.

Ovary and Corpus Luteum: Slide # 8-10 (H&E and Trichrome)

- In low magnification image, the ovary is lined by a simple cuboidal mesothelium, the **germinal epithelium**. Beneath the mesothelium is a layer of fibrous tissue, the **tunica albuginea**. Underlying the tunica albuginea is the less fibrous stroma and scattered within it are ovarian follicles. These structures constitute the ovarian cortex. In the center of the ovary is the medulla composed of loose connective tissue with large blood vessels and lymphatics.
- Ovarian follicles (found only in the cortex) are composed of an oocyte encircled by epithelia-like follicular (granulosa) cells. Be aware that all follicles are not cut through the center; many sections just graze the surfaces of follicles so that their stages may be difficult to distinguish.
- **Primordial follicles** consist of flat follicular cells surrounding a primary oocyte.
- **Primary follicles** have one (unilaminar) or more (multi laminar) layers of cuboidal follicular cells surrounding a primary oocyte. A thin **zona pellucida** (made of glycoprotein and proteoglycans) lies between the oocyte and the follicular cells.
- **Secondary (antral) follicles** have a thick granulosa (follicle cell layer) and a zona pellucida. They are distinguished by an **antrum** filled with liquor folliculi. The hillock of follicle cells that attach the oocyte to the follicle wall is called the **cumulus oophorus**. Around the periphery of the follicle are stroma cells which comprise the **theca interna**. The flattened spindle-shaped cells of the **theca externa** may also be apparent. This layer can be less distinctly visible.
- **Tertiary (Graafian / mature) follicles** have a large antrum, and a thin granulosa layer. The mound-like **cumulus oophorus** will attach the zona pellucida - encapsulated oocyte to the follicle wall. A single layer of follicle cells that form a ring around the surface of the oocyte (still primary oocyte) are called the **corona radiata**. Both layers of the theca are much more evident.
- **Atretic follicles** are degenerating follicles and may arise during any developmental stage.

- The **corpus luteum** is a large group of yellowish cells from the wall of a tertiary follicle after the oocyte is ovulated. It is comprised mainly of **granulosa lutein** cells derived from the granulosa of the follicle and a smaller layer of **theca lutein** cells derived from the former theca interna. It serves as a temporary endocrine gland which persists for about 2 weeks. If fertilization and implantation occur, it persists for several months. The **corpus albicans** is white, fibrous scar tissue that remains in the ovary after the corpus luteum has degenerated.

Identify and check-off each of the following:

☐ Germinal epithelium	☐ Tunica albuginea	☐ Cortex	☐ Medulla
☐ Primordial follicle	☐ Primary follicle	☐ Secondary follicle	
☐ Tertiary follicle	☐ Granulosa cell	☐ Theca Interna	☐ Antrum
☐ Zona pellucida	☐ Cumulus oophorus	☐ Corpus luteum	☐
☐ Corpus albicans	☐ atretic follicle		

Corpus Luteum/Luteal gland: Slide #10

- At ovulation, the wall of the follicle, made up of the remaining granulosa and thecal cells is thrown into deep folds as the follicle collapses on itself. The follicle is now transformed into the corpus luteum (yellow body) or luteal gland.
- The corpus hemorrhagicum is initially formed. Capillaries from the theca interna bleed into the lumen of the follicle, forming the corpus hemorrhagicum with a central clot.
- The corpus luteum then undergoes the process of luteinization whereby the granulosa and theca interna cells differentiate into granulosa luteal and theca luteal cells. Microscopically, the luteal cells display features of steroid-producing cells (abundant mitochondria with tubular cristae). The granulosa lutein cells produce estrogen, progesterone and inhibin – controls the production and secretion of FSH from the anterior pituitary gland. Theca lutein cells secrete androgens and progesterone. These hormones stimulate the growth and secretory activity of the uterine lining.
- If there is no fertilization, the corpus luteum remains active for an average of 10 days (corpus luteum of menstruation). If no hCG then the rate of secretion of estrogen and progesterone declines and the corpus luteum begins to degenerate. This later gives rise to the corpus albicans, which slowly disappears over a couple months.

Identify:

☐ Corpus luteum
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Uterus: Slide # 11-12 (H&E and diagram)

The wall of the uterus has 3 structurally and functionally distinct layers:

- 1) **Perimetrium** is the outermost layer composed of dense irregular connective tissue.
 - 2) **Myometrium** is the middle and thickest layer comprised of interwoven layers of smooth muscle. Cyclic changes in cell numbers and size are mediated by the estrogen / progesterone balance. The myometrium is responsible for muscular support and for uterine contractions during parturition.
 - 3) **Endometrium** is analogous to the mucosa and is lined by simple columnar epithelium with many ciliated cells and simple tubular glands rich in mucus secreting cells. The epithelium is underlain by a lamina propria. The endometrium is divided into a basal layer adjacent to the myometrium and a functional layer. The functional layer varies in thickness with stages of the menstrual cycle and is sloughed off during menstruation.
- Large blood vessels supply the uterus. Branches called **straight arteries** supply the basal layer of the endometrium and these in turn branch to form **spiral arteries** which supply the functional layer. Spiral arteries constrict when progesterone / estrogen levels drop prior to menstruation. This results in lack of blood flow and leads to ischemia and shedding of the functional layer during menstruation.
 - Both the uterus and its stage in the menstrual cycle are identified by examining the endometrium. The **proliferative phase** has a relatively thin endometrium with short, straight glands. The **secretory phase** is characterized by a thick endometrium with noticeably elongated, widened, and coiled glands.
 - Secretions may be visible in the uterine lumen and in the apical vacuoles of epithelial cells. The menstrual phase is characterized by stagnant pools of blood in distended vessels.

Identify and check-off each of the following:

☐ **Endometrium** ☐ **Myometrium** ☐ **Perimetrium** ☐ **Lamina propria**
☐ **Basal layer** ☐ **Functional layer** ☐ **Simple columnar epithelium** ☐
☐ **Uterine glands** ☐ **Proliferative stage** ☐ **Secretory stage**

Placenta: Slide # 14 (H&E)

- As the zygote implants in the endometrium, its trophoblast cells proliferate, and the placenta becomes a syncytium of **placental villi** interdigitated with the **intervillous space** containing maternal blood (supplied by the spiral arteries of the uterus). The placental villi project into the intervillous space and facilitate waste and nutrient exchange between the embryo and the mother. The villi contain a core of connective tissue overlain by cytotrophoblast cells. Some villi form large trunks which serve as a base for the smaller, more numerous villi involved with exchange. Villi that penetrate the syncytium and bind directly to uterine tissue are called anchoring villi.

Identify and check-off each of the following:

☐ Placental villi	☐ Intervillous space	☐ Cytotrophoblast cells
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Mammary Gland (Breast): Slide # 15-16 (H&E and Electron Micrograph)

- At low magnification, the section is primarily composed of dense irregular connective tissue, islands of inactive lactiferous ducts, clusters of unilocular adipose tissue, and blood vessels.
- Scattered throughout the field are islands of **intralobular lactiferous ducts**. These are within lobules and have few secretory or alveolar units. The intralobular ducts are lined by simple cuboidal epithelium, but the epithelium may look stratified due to the presence of **myoepithelial cells** along the basal aspect. Myoepithelial cell nuclei contain condensed chromatin, and are triangular compared to the pale, round nuclei of the cuboidal cells. They may be difficult to discern on your slide.

Identify and check-off each of the following:

☐ Lactiferous ducts	☐ Simple cuboidal epithelium	☐ Myoepithelial cells
☐ Adipose tissue	☐ Dense irregular connective tissue	

Lactating Mammary Gland (Breast)

- At low magnification, the epithelium of the ducts is readily distinguished from the connective tissue. Lobules and large **intralobular lactiferous ducts** are surrounded by connective tissue.
- Each lobule has numerous distended alveoli that have proliferated along the length of the intralobular duct that contain lipid, protein and polysaccharide secretion products in their lumens. Intralobular ducts are smaller branches of the lactiferous ducts. **Secretory epithelial cells** are relatively large, slightly basophilic and have a vacuolated cytoplasm. Plasma cells are may be seen in the vascular lamina propria within the lobules. Flattened **myoepithelial cells** can also be observed beneath the secretory cells.

Identify and check-off each of the following:

☐ Secretory alveoli	☐ Lactiferous ducts	☐ Secretory cells
☐ Myoepithelial cells		

Study Questions:

What is the function of myoepithelial cells?

What mode(s) of secretion are used by the glandular tissue of the breast?

List the order of development of ovarian follicles.

What is the role of glycogen release in the vagina?

List the two types of epithelia found at the cervico-vaginal junction.

Which layer of the endometrium is lost during menstruation?

Which of the ovarian follicles does not contain a zona pellucida?

List the epithelia found in the oviducts, in the uterus, in the cervix and in the vagina.

What is found in the intervillous space of the placenta?